

Technical Description

ENERCON SCADA

PDI-OPC

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1 Introduction

The ENERCON SCADA PDI-OPC is an OPC XML-DA server in accordance with the OPC Foundation specifications V1.01. The ENERCON SCADA PDI-OPC allows for the online evaluation and control of individual wind energy converters (WECs), wind farms and other system independent of the ENERCON SCADA Remote software.



Due to potential communication problems or disruptions, it is not possible to guarantee fault-free, complete data transmission to the WECs in every case. The start/stop function is subject to this restriction.

Purpose of this document

This document describes the use of the ENERCON SCADA PDI-OPC. This document does not describe the basics of the OPC standard. For a detailed description of the data tree and the items used in OPC, please refer to the “Datenblatt ENERCON SCADA PDI-OPC” (Data Sheet ENERCON SCADA PDI-OPC) (D0138302).

User skills

A basic knowledge of the OPC standard is required. Information about the OPC standard can be found at www.opcfoundation.org.

2 ENERCON SCADA PDI-OPC within the ENERCON SCADA system

ENERCON SCADA PDI-OPC is implemented as a software solution on the ENERCON SCADA Server and expands its functionality to include that of an OPC XML DA server. ENERCON SCADA PDI-OPC enables online access to wind farm and WEC data, independent of the ENERCON SCADA Remote software. In addition, setpoints can be transmitted.

The OPC XML DA protocol (V1.01) used for that purpose is a non-proprietary, Internet-capable data exchange protocol. Thanks to its simple structure, it is particularly suited for exchanging data located on different networks. Data can also be transmitted over the Internet.

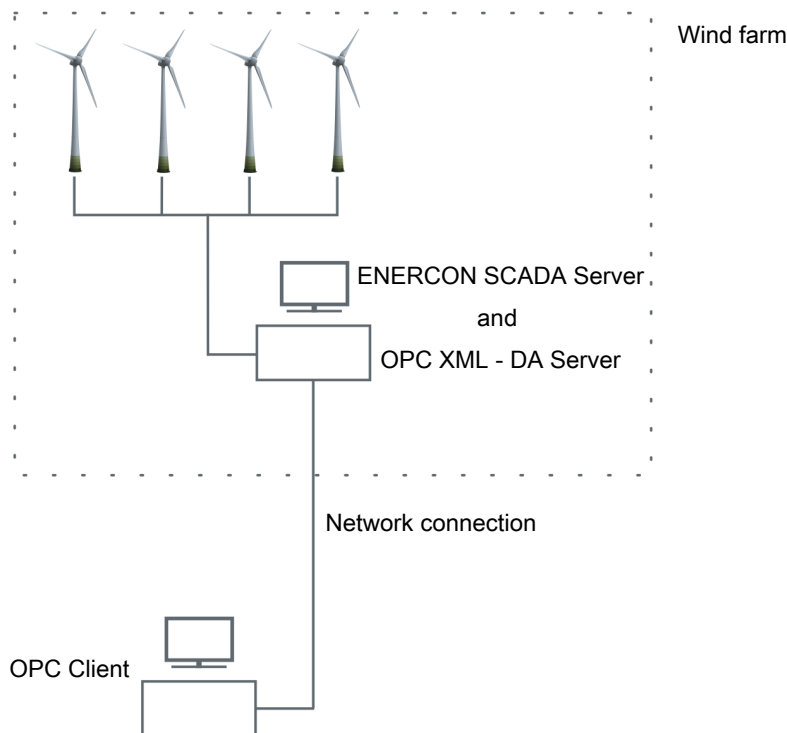


Fig. 1: ENERCON SCADA PDI-OPC within the ENERCON SCADA system

The OPC XML DA server supports data buffering for clients (Basic Polled Subscription Interaction; see ch. 3.2, p. 6). Once a client activates this function, the server temporarily stores changes to send them at the client's next request.

Communication failures may cause data inconsistency. To ensure that key data (e.g. status data) can be retrieved from the OPC server in their entirety, the data history is made available for synchronisation purposes. Related data such as status data (status; substatus; information, etc.) are provided as an array, since they are generated at the same point in time and are assigned identical timestamps.

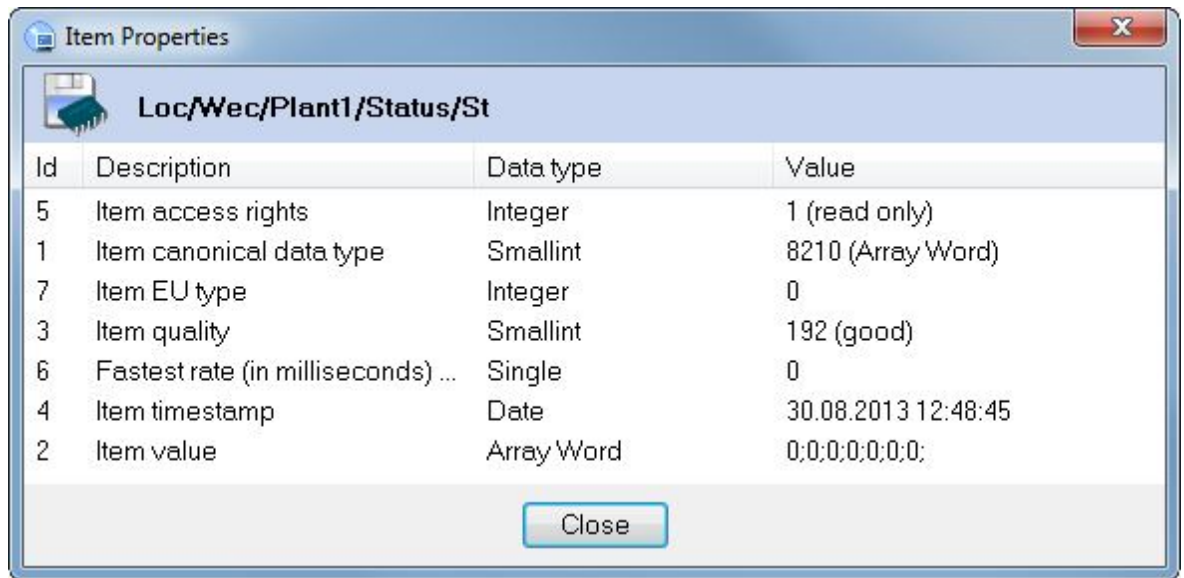
3 Functional description

3.1 Data structure

3.1.1 Data management

All data sources are managed in the namespace of the OPC server. With the ENERCON SCADA PDI-OPC, the namespace has been set up as a tree structure. Nodes serve as structuring elements of this tree. Items form the data proper which the client can access in read-only and/or write mode.

Each item identified in the namespace by a unique name (here: Loc\Wec\CalcP) is described by the following item properties:



Id	Description	Data type	Value
5	Item access rights	Integer	1 (read only)
1	Item canonical data type	Smallint	8210 (Array Word)
7	Item EU type	Integer	0
3	Item quality	Smallint	192 (good)
6	Fastest rate (in milliseconds) ...	Single	0
4	Item timestamp	Date	30.08.2013 12:48:45
2	Item value	Array Word	0;0;0;0;0;0;0;

Fig. 2: Data record of an online value and its item properties

Tab. 1: Data record of an online value and its item properties

ID	Parameter	Description
5	Item access rights	Indicates whether the item is read-only or also writable.
1	Item canonical data type	Indicates the data type of the item.
7	Item EU type	Indicates if the value is an enumerated type (defined states of a group of values).
3	Item Quality	Indicates the quality of the item (192 (good)/28 (uncertain)).
6	Fastest rate	Indicates how quickly the value can be updated on the OPC server. If this is zero, the value may change randomly (event-based value).
4	Item timestamp	Indicates when the item was written (based on the time zone (UTC) configured on the server).
2	Item Value	Indicates the value of the item in the form of the data type.



The Fastest Rate parameter applies only to online data. Since report, status, availability and control data are event-based, Fastest Rate is set to zero for these items. If the WEC is inactive or if communication is disrupted, the value for Item Quality will be set to uncertain. For report data, the respective value will be set to uncertain if data points are not populated with data.

3.1.2 Data types

The OPC server provides the following data types:

Online values

Online values are instantaneous values that the OPC server refreshes as often as permitted by the wind farm's ENERCON SCADA System. Their informative value is limited so that there is no need to store them on the ENERCON SCADA Server.

The data refresh rate depends on the number of wind energy converters installed in the wind farm, the structure of the wind farm data bus, and – primarily – the communication hardware selected. The structure of each specific project defines the highest possible transfer rate. It is communicated by the system and can be viewed on the client. The shortest refresh interval within ENERCON SCADA PDI-OPC is one second.

Fixed parameters such as the serial number of a specific wind energy converter also belong to the online values group.

Report data

Report data are generated by various measurements (data loggers) as part of the wind energy converter monitoring process. These include wind energy converter status data as well as average values for a specific time period. The latter include data collected by the ENERCON SCADA Server.

The following averages are commonly computed: one minute, ten minutes, one day, one week, one month. All of these are based on one-minute averages. All 10-minute values (and longer intervals) are stored on the ENERCON SCADA Server. The system stores

historical report data. The OPC server provides not only current values but also values from past time intervals. This ensures the consistency of the data on the ENERCON SCADA Server and the customer's PC even if online communication is poor.



If the ENERCON SCADA Server cannot poll any data from the wind energy converters and devices, it will not be able to compute any 10-minute averages. Therefore, no report data will be available in OPC for that period.

Control data

Control data can be used by the client to change wind farm or wind energy converter parameters. This includes open-loop control of the wind farm control system as well as starting and stopping individual WECs.

3.2 Data querying by means of Basic Polled Subscription Interaction

In accordance with the OPC XML DA V1.01 specification, data are queried using Basic Polled Subscription Interaction. This is a method for exchanging data between server and client software. The diagram below illustrates the principle:

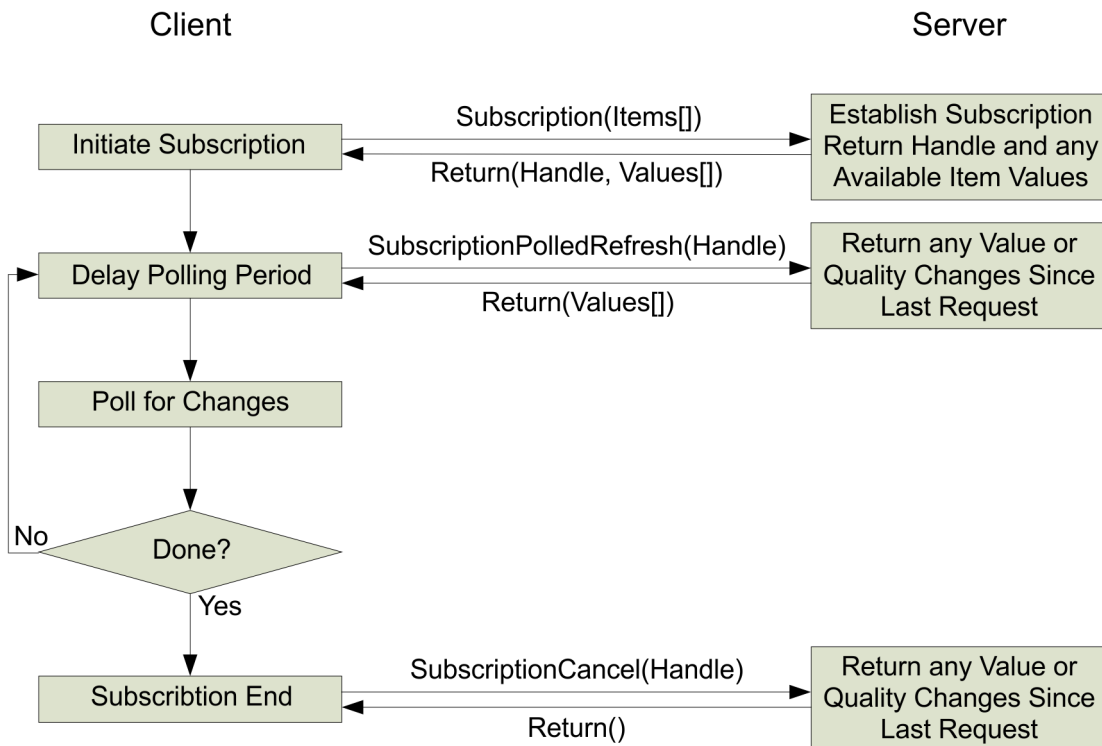


Fig. 3: Basic Polled Subscription diagram

The client software initialises a subscription with the data points to be requested on the server, which are polled cyclically by the client. During the Delay Polling Period, the server records the changes to the requested data points in order to make them available to the client software during the Polling Refresh. Using this form of data compression, only the changed data points are communicated to the client. In order to terminate a subscription, the client must notify the server. Each installed subscription reserves resources on the server in the form of a dedicated thread.



A maximum of 100 subscriptions is permitted so as not to compromise the operational readiness of the ENERCON SCADA PDI-OPC. The server will close any subscription whose data have not been collected after the polling period has expired (Subscription Ping Rate).

3.3 Namespace in the ENERCON SCADA PDI-OPC

The namespace describes the structure of the data points in the ENERCON SCADA PDI-OPC. Whereas the structure of the transmitted data is specified by ENERCON, their graphical representation on the client depends on the software used by the customer for that purpose. Software programs that represent the nodes as file folders forming a tree structure are commonly used. Items in these folders are shown as files.

The screenshots used in this document were made using Kassl OPC Explorer 1.36.

There are three nodes located in the root folder of the ENERCON SCADA PDI-OPC:

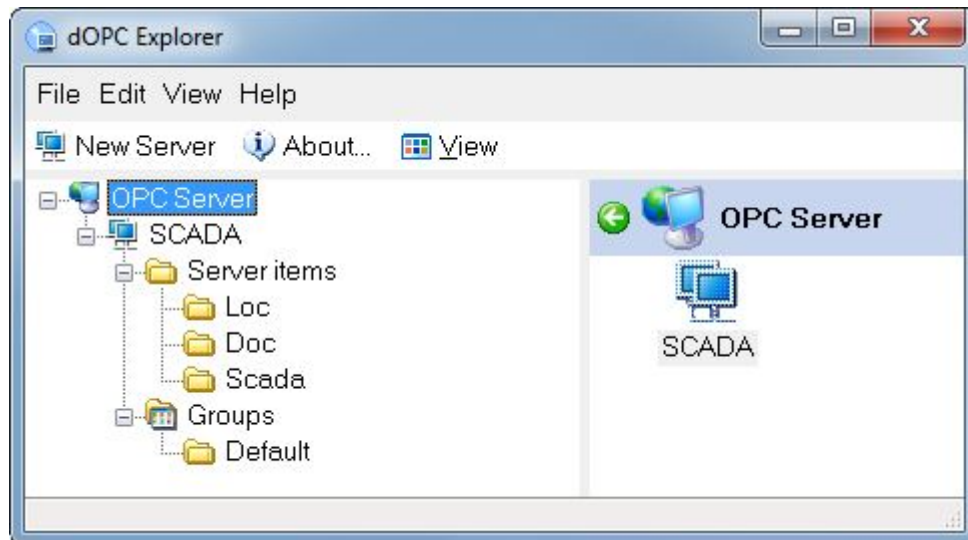


Fig. 4: Nodes in the root folder of the ENERCON SCADA PDI-OPC namespace

Tab. 2: Root node nodes

Node	Description
Loc	All plant data of a wind farm (e.g. data from wind energy converters; open-loop and closed-loop control systems such as RTU or FCU; METEO weather measurement data).
Doc	Short description of the items in ENERCON SCADA PDI-OPC. A short description of the items is provided based on the data structure of the Loc folder.
Scada	SCADA-specific items such as open-loop control/closed-loop control/time.

3.3.1 The Location node (Loc)

The Location label refers to a wind farm. All information referring to this location is available under the Loc node. Data sources include not only the wind farm WECs but all other components of the ENERCON SCADA system such as ENERCON METEO, ENERCON SCADA RTU, and similar systems.

For a detailed description of the nodes and items in the Location path, please refer to the “Datenblatt ENERCON SCADA PDI-OPC” (Data Sheet ENERCON SCADA PDI-OPC) (D0138302).

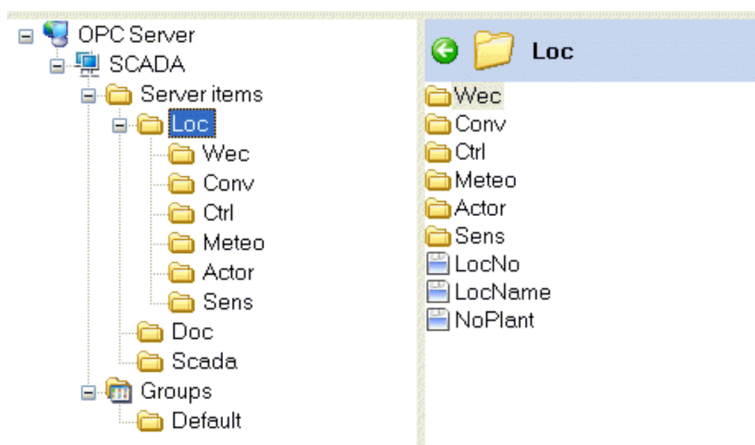


Fig. 5: Location node (Loc)

Tab. 3: Location node (Loc)

Node/item	Description
Wec	Data from the wind energy converters
Sens	Data from the sensor systems (e.g. medium-voltage data acquisition)
Conv	Data from the converter systems (e.g. solar converter)
Ctrl	Data from the open-loop control systems (e.g. ENERCON SCADA RTU)
Actor	Data from the actors (e.g. ENERCON SCADA PDI)
Meteo	Data from the ENERCON METEO weather recording system
LocNo	Wind farm number assigned by ENERCON; same as the ID used for identification by the ENERCON SCADA Remote software
LocName	Unique name of the wind farm defined by the customer
NoPlant	Number of all plants in this wind farm, including all devices connected to the ENERCON SCADA System

3.3.2 The Documentation node (Doc)

Besides the Location path, users have online access to the Documentation path. The items in this path provide explanations of the corresponding items in the Location path. Also, this can be used to look up abbreviations used for nodes and items in ENERCON SCADA PDI-OPC.

Id	Description	Data type	Value
5	Item access rights	Integer	1 (read only)
1	Item canonical data type	Smallint	8 (OleStr)
7	Item EU type	Integer	0
3	Item quality	Smallint	192 (good)
6	Fastest rate (in milliseco...	Single	0
4	Item timestamp	Date	30.08.2013 12:48:54
2	Item value	OleStr	No. of historical data

Fig. 6: Explanation of corresponding NoSt item in the Location path

The structure of the Documentation path is largely identical to that of the Location path. The <grp> (group) node combines all wind energy converters and the ENERCON SCADA system devices of a wind farm. The total number is again reflected in the *NoPlant* item.

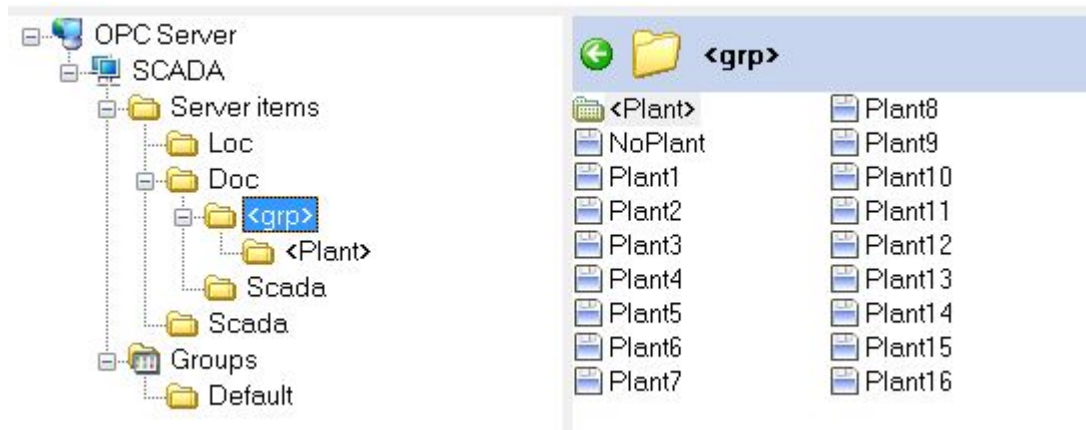


Fig. 7: Documentation node (Doc)

3.3.3 The Scada node

The Scada node contains SCADA-specific items resulting from the project-specific features of the ENERCON SCADA system.

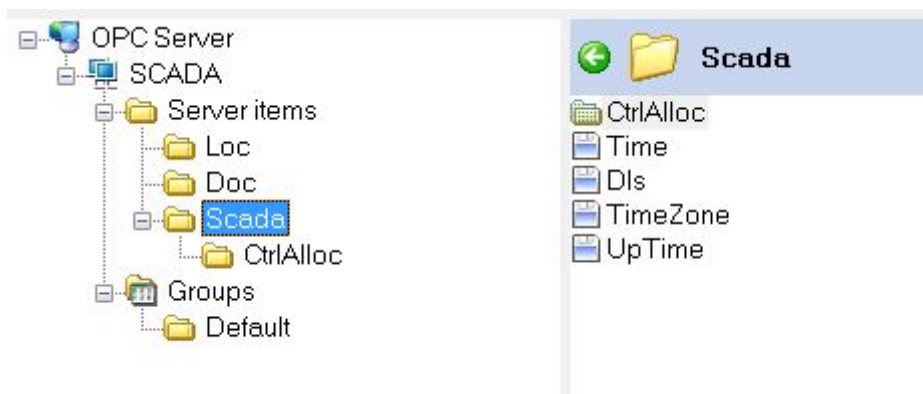


Fig. 8: Scada node (example)

Tab. 4: Scada node

Node/item	Description
CtrlAlloc (example)	Items for global allocation of all access for control purposes
Time	Depending on the configuration, the following times are transmitted: ■ UTC activated The value transmitted is the coordinated universal time (UTC; format dd.mm.yyyy hh.mm.ss). ■ UTC deactivated The value transmitted is the local server time (format dd.mm.yyyy hh.mm.ss).
DIs	Flag for daylight saving time.
TimeZone	Time zone east of the prime meridian in minutes (German time zone = +60)
UpTime	Start-up time of the OPC server, e.g. after a grid fault. This point in time is not identical with the time of installation.

3.4 Transmission of setpoints and control values

Both for open-loop and closed-loop control processes, setpoints or control values may need to be transmitted to the wind farm. The ENERCON SCADA PDI-OPC supports the transmission of setpoints and control values for example for starting and stopping a wind energy converter or for providing the wind farm controller (e.g. ENERCON SCADA RTU and ENERCON SCADA FCU) with setpoints.

ENERCON SCADA PDI-OPC processes these setpoints and control values in sessions. This ensures that actions are identifiable and that only one client at a time can perform an action.

This mechanism is merely intended to manage and identify access by trusted communication partners. It is not a security feature such as VPN or SSL encryption.

```

graph TD
    subgraph Client
        C1{Check session status free?}
        C2[Initiate session ID and private key random value]
        C3[Write SessionRequest: Session ID, user ID, private Key]
        C4{Check session status and session ID}
        C5[Read public key]
        C6[Write SetCtrl 1: Parameter 1, private key, public key]
        C7[Write Set Ctrl<n>: Parameter <n>, private key, public key]
        C8{Check session status and session ID}
        C9[Write SessionSubmit: Private key, public key]
        C10{Check session status and session parameter}
        C11{Loop Parameter?}
        C12[Session status: Await session end, starting timeout]
    end

    subgraph Server
        S1{Check session status free?}
        S2{Check user ID?}
        S3[Session status: Login was successful, start session timeout]
        S4{Check private key and public key}
        S5{Maximum of incorrect entries reached?}
        S6[Transfer parameter to OPC]
        S7[Session status: Parameters entered, reset session timeout]
        S8{Check private key and public key}
        S9[Set parameters in SCADA]
        S10{Check loop mode?}
        S11[Session status: Await session end, starting timeout]
        S12[Session status: error Start error timeout]
        S13[Await end of timeout]
        S14[Start session timeout]
        S15[Session status: Waiting period in loop mode]
        S16[Await end of session timeout]
    end

    C1 -- No --> C1
    C1 -- Yes --> C2 --> C3 --> S1
    S1 -- No --> C1
    S1 -- Yes --> S2
    S2 -- No --> S1
    S2 -- Yes --> S3 --> C4
    C4 -- No --> C1
    C4 -- Yes --> C5 --> C6 --> S4
    S4 -- No --> S5
    S4 -- Yes --> S6 --> S7
    S7 --> S8
    S8 -- No --> S5
    S8 -- Yes --> S9 --> S10
    S10 -- Yes --> S15 --> S13 --> S14 --> S3
    S10 -- No --> S11
    S11 --> C10
    C10 -- No --> C4
    C10 -- Yes --> C9 --> S8
    S5 -- Yes --> S12 --> S13 --> S14 --> S3
    S5 -- No --> S1
    C12 --> S16
    S16 --> S1
  
```

The flowchart illustrates the OPC UA Session Establishment and Loop Mode process, divided into Client and Server actions.

Client Actions:

- Check session status free? (Decision)
- Initiate session ID and private key (random value)
- Write SessionRequest: Session ID, user ID, private Key
- Check session status and session ID (Decision)
- Read public key
- Write SetCtrl 1: Parameter 1, private key, public key
- Write Set Ctrl<n>: Parameter <n>, private key, public key
- Check session status and session ID (Decision)
- Write SessionSubmit: Private key, public key
- Check session status and session parameter (Decision)
- Loop Parameter? (Decision)
- Session status: Await session end, starting timeout

Server Actions:

- Check session status free? (Decision)
- Check user ID? (Decision)
- Session status: Login was successful, start session timeout
- Check private key and public key (Decision)
- Maximum of incorrect entries reached? (Decision)
- Transfer parameter to OPC
- Session status: Parameters entered, reset session timeout
- Check private key and public key (Decision)
- Set parameters in SCADA
- Check loop mode? (Decision)
- Session status: Await session end, starting timeout
- Session status: error Start error timeout
- Await end of timeout
- Start session timeout
- Session status: Waiting period in loop mode
- Await end of session timeout

Flow Details:

- The Client initiates the session by checking if the session status is free. If not, it loops back. If yes, it initiates a session ID and private key, then writes the SessionRequest to the Server.
- The Server checks the session status. If not free, it loops back. If free, it checks the user ID. If not, it loops back. If yes, it sets the session status to "Login was successful, start session timeout" and sends the SessionRequest to the Client.
- The Client checks the session status and session ID. If not, it loops back. If yes, it reads the public key and writes SetCtrl 1 to the Server.
- The Server checks the private key and public key. If not, it checks if the maximum of incorrect entries has been reached. If yes, it sets the session status to "error Start error timeout". If no, it transfers the parameter to OPC.
- The Client writes Set Ctrl<n> to the Server.
- The Server sets the session status to "Parameters entered, reset session timeout" and checks the private key and public key again. If not, it checks for the maximum of incorrect entries. If yes, it sets the session status to "error Start error timeout". If no, it sets parameters in SCADA.
- The Client checks the session status and session ID. If not, it loops back. If yes, it writes SessionSubmit to the Server.
- The Server checks the loop mode. If yes, it sets the session status to "Waiting period in loop mode" and starts the session timeout. If no, it sets the session status to "Await session end, starting timeout" and sends the SessionSubmit to the Client.
- The Client checks the session status and session parameter. If not, it loops back. If yes, it checks the Loop Parameter. If yes, it loops back. If no, it sets the session status to "Await session end, starting timeout".
- The Server awaits the end of the session timeout and then checks the session status again.

Fig. 9: OPC session diagram

The simplified diagram illustrates four process steps for transmitting parameters or control values to the OPC server.

- A session is reserved for access by specifying a session ID, a user ID, and a private key. The respective values are written to an array.
- The appropriate parameters are entered into the items. They do not take effect yet.
- Parameters become active once the system has acknowledged their transfer.
- The timeout must have elapsed before a new session can be reserved.

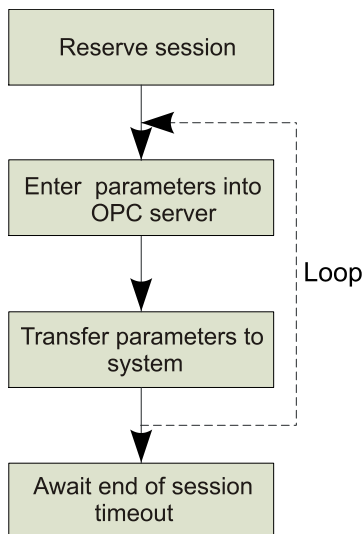
There are two types of sessions:

Single Session

As described above, a single session comprises four steps. It is typically used to set parameters or control values.

Loop Session

A loop session is used to receive cyclic data from clients. This forces the continuous communication with the clients and requests Alive signals from the clients. At first, this session progresses just like a single session. However, once the parameters have been entered into the system (Session Submit), the session will return to parameter input without making another reservation. Typically, one Session Submit is enough to execute the Alive signal.



In principle, different parties can transmit external setpoints. For this reason, the relevant permissions need to be defined for each individual project and must be integrated into the ENERCON SCADA System. The definition of the permissions must be agreed in advance with ENERCON Service.

For security reasons, authentication of the sender's identity is required for each setpoint transmission.

ENERCON offers setpoints for the following functions:

- Switching a wind energy converter on or off (start/stop)
- Setpoint signal to wind farm controller

The duration of a session is defined by the regular timeout. Each time a setpoint or parameter has been modified, the extended timeout starts that prevents another change for a specific time period. This dynamic interval ensures that electrical values have been fully adjusted before the next setpoint modification takes place. In the event of an incorrect input, another input will be prevented for a specified time period.

Timings

Regular timeout during a session	Configurable (minutes)
Extended timeout after setting a parameter/control value	Configurable (minutes)
Session end timeout (delay for new reservation)	10 s
Error timeout (after three incorrect inputs)	300 s
User ID error timeout	300 s

3.5 OPC session workflow

The following section explains the workflow of an OPC session, using the transmission of a start/stop setpoint for a wind energy converter as an example.

3.5.1 Session start

In order to start a session, the SessionRequest item must be written on the OPC server. This item can be found in the Ctrl node of the respective wind energy converter.

The following values must be written to the array in the sequence defined below:

Tab. 5: Description of session items

Session ID	Random value from the client which the client can use to determine whether a reservation was successful. In the example shown in the illustration below, the value of the Session ID is 11. This value can be read by the clients.
User ID	The user ID represents a user in the ENERCON SCADA System. SCADA users and their permissions are created in the ENERCON SCADA System by ENERCON Service IT. This value cannot be read and is therefore invisible to other clients.
Private Key	The Private Key is a random value and a subkey for access to the respective session. This value cannot be read and is therefore invisible to other clients.

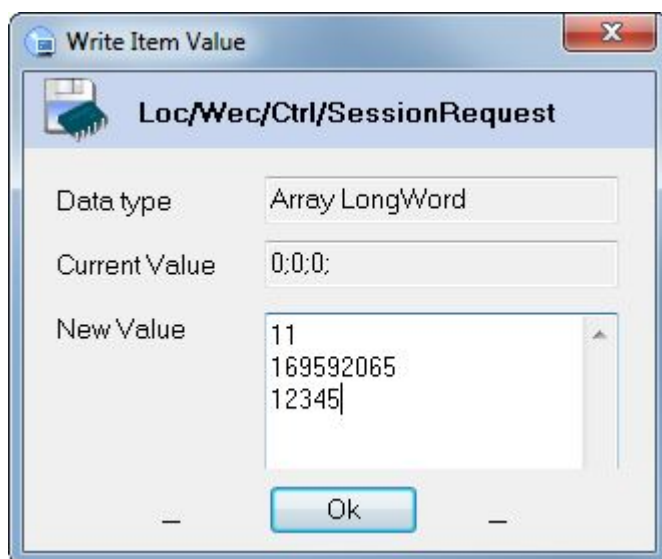


Fig. 10: Writing the SessionRequest item

Once the values for Session ID, User ID and Private Key have been written to the SessionRequest item, the session state (which can be viewed in the Session State item) changes to 1 – Session reserved if the login was successful. Otherwise, an error code appears and a timer starts to prevent any other login attempt for a specific period. This time period is defined by ENERCON.

Tab. 6: Feedback code

Feedback code	Description
0	Session free
1	Session reserved
2	Parameter input
3	Wait time in loop mode
4	Session end wait time
5	Session blocked (global reservation)
108	Busy
109	Access denied
121	Illegal value
174	Incorrect user ID
175	Insufficient access rights

If the login was successful, a session timeout is initiated and a public key for the session is written to the SessionPubKey item.

All operations carried out during this session use the private key and the public key. To confirm that this is the user who has actually reserved this session, the user must verify their session ID.

3.5.2 Input of control values

The control values for a wind energy converter are defined in the SetCtrl item. The following values must be written in the sequence defined below:

Control Value	One of the following control values must be used for starting or stopping the wind energy converter. 0 = WEC start 1 = WEC stop at 60° blade angle 2 = WEC stop at 90° blade angle 3 = WEC gradient stop at 60° blade angle 4 = WEC gradient stop at 90° blade angle 5 = Ice detection stop 6 = Shadow casting stop 7 = Animal protection stop at 60° blade angle 8 = Animal protection stop at 90° blade angle
Private Key	The private key selected by the user at the start of the session
Public Key	The public key made available by the server at the start of the session

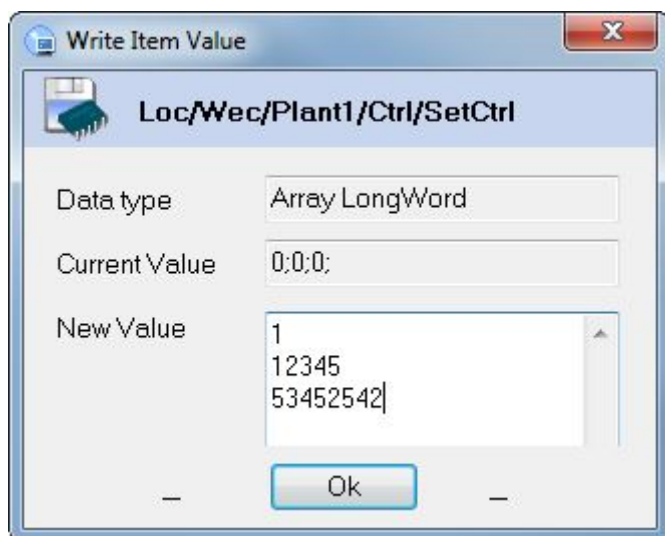


Fig. 11: Writing the SetCtrl item

The private key and the public key must match the keys used at the start of the session. If the user enters one of these keys incorrectly three times, the corresponding error code will be displayed in the SessionState item and a timeout will be started. This timeout prevents the input of new data for a specific time period.

Once the data have been keyed in correctly, the SessionState will change from Session-Reserved to SessionParameterEntered. The timeout for that session will also be reset so that the user will still be able to edit the values.

3.5.3 Data transmission to the ENERCON SCADA system

After the values have been entered into the OPC client, they must be transmitted to the OPC server in the ENERCON SCADA system for execution. This process is started by writing the private key and the public key to the SessionSubmit item.

The private key and the public key must match the keys used at the start of the session. If the user enters one of these keys incorrectly three times, the corresponding error code will be displayed in the SessionState item and a timeout will be started. This timeout prevents the input of new data for a specific time period.

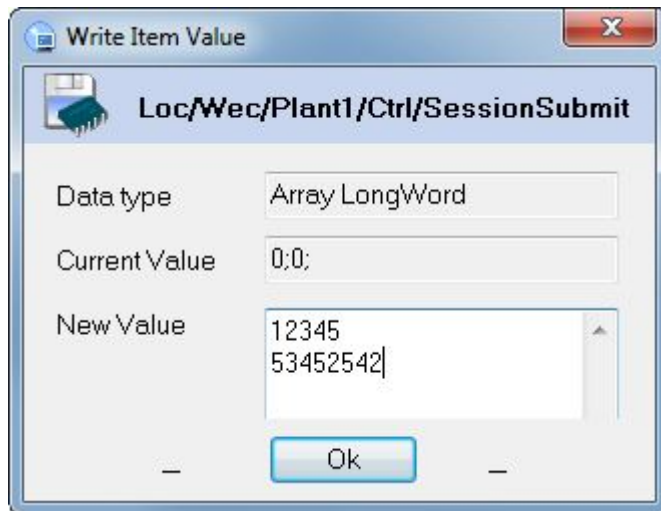


Fig. 12: Writing the SessionSubmit item

After the private key and the public key have been transmitted, the session state changes to Wait Session End (4). After the status change a timer will be started to prevent the user from entering more data. If the wind energy converter has accepted the modified control data, the Ctrl item will show the user the plant.

Tab. 7: Feedback code

Feedback code	Description
0	WEC start
1	WEC stop at 60°
2	WEC stop at 90°
121	Wrong value
129	WEC stopped at 60° (with higher priority). Cannot start.
130	WEC stopped at 90° (with higher priority). Cannot start.
255	Communication error. A timeout has occurred.

3.6 Global reservation of all access for control purposes

The global reservation function can be used to reserve all access for control purposes by means of ENERCON SCADA PDI-OPC for a single user (user ID). Following initial reservation of the session (SessionRequest), there is merely a cyclic run of the session (SessionSubmit) as a loop session.

The reservation is associated with a priority assigned to each ENERCON SCADA PDI-OPC user (each user ID) in the ENERCON SCADA System. Level 0 has the highest priority.

Users with higher priority levels can cancel the existing reservations of users with lower priority levels and can thus take control of the access for control purposes in the ENERCON SCADA PDI-OPC.

Following successful reservation, all SessionState items will display this reservation indicating the session ID and the public key. Access for control purposes by means of these private and public keys is now possible without having to make the respective reservation in advance.

Data are transmitted in a loop session. The UserName item (login name of the user in the ENERCON SCADA system) resides in the Scada/CtrlAlloc node.

3.7 Processing of ENERCON PDI-OPC data by the customer

The transmission of the data provided by the ENERCON SCADA Server (= OPC XML DA server) at the wind farm to the customer's IT system is the responsibility of the customer. ENERCON Service IT offers support in selecting the appropriate data link.

The customer's IT system must include a software that is designed for data exchange using the OPC XML DA protocol. A wide range of commercial software applications is available. Customers can decide individually which data to select, view and store on their systems.

The OPC XML DA server in the wind farm is designed so that customers can save to their local systems the exact same data that exist in the wind farm, even if online communication is poor. This is achieved by providing the entire history in addition to current online and average values. In this way, if data transmission is interrupted, customers can fill in any gaps in their data at a later stage. However, this process must be initiated by the OPC client (=customer) because the server cannot determine which gaps may have occurred in the customer's data.

Because a significant portion of the communication and data processing takes place outside the wind farm and thus beyond ENERCON's responsibility, customers have no legal claim to complete data transmission.

3.8 Fail-safety and default values

If the setpoint signalling via the ENERCON SCADA PDI-61400 interface of the ENERCON SCADA Server is disrupted, the ENERCON SCADA Server cannot receive any new setpoints via the interface. Setpoint signalling may be disrupted by the following events, for example:

- Interface errors
- Disrupted communication with signal source
- Server off-line

Interface malfunction

If communication with the interface or the interface itself is faulty and the server is online, the controller used (ENERCON SCADA FCU or RTU) will continue closed-loop control using the last setpoints received.

Server offline - behaviour of FCU

If control is via the ENERCON SCADA FCU and the ENERCON SCADA Server is shut down due to a fault for instance (server offline), the FCU will continue operation using the last setpoints received.

Server offline - behaviour of RTU

If control is via the ENERCON SCADA RTU and the ENERCON SCADA Server is shut down due to a fault for example (server offline), control via the RTU will not be possible as long as the server is down.

If parameter 1308 *SCADA farm control* (wind energy converter parameter) is activated, the ENERCON SCADA Server will save the last valid control value to the following wind energy converter parameters:

- 1209 *Dynamic reactive power (Q/P) (closed-loop wind farm control fault)*
- 1210 *Static reactive power (closed-loop wind farm control fault)*
- 1400 *Maximum power (closed-loop wind farm control fault)*

To ensure any slight changes to RTU control values do not result in permanent changes to the last valid control values stored in the wind energy converters, the control values are not written until a specified deadband has been exceeded/fallen short of. The following values apply here:

- P: $\pm 10.01\%$
- Q: $\pm 10.01\%$
- Q/P: ± 0.170 (9.65°)

If wind energy converter parameter 1308 is deactivated, the wind energy converters use the control values stored in parameters 1209, 1210 and 1400. The control values are not stored by the ENERCON SCADA Server if parameter 1308 is deactivated. The control values stored will then be the default values configured.

Checking the setpoint

An evaluation of whether a setpoint has been received correctly and correctly processed by the FCU or RTU can be made via the feedback data points for the FCU/RTU interfaces for the relevant setpoint.



Checking the correct receipt and transfer of a setpoint is the job of the entity which specified the setpoint (e.g. the operator/owner).

4 System requirements

Hardware requirements

ENERCON requires the following hardware platform:

- Industrial PC
- P4 or equivalent CPU; 3 GHz
- ≥ 2 GB RAM
- Temperature range between $-10\text{ }^{\circ}\text{C}$ and $+40\text{ }^{\circ}\text{C}$ as per IEC 61400-1
- RJ45 network connector



Setup of the ENERCON SCADA PDI-OPC server function in new wind farms or as a modification to existing systems can only be carried out by ENERCON. All hardware must be approved by ENERCON!

Network configuration

To ensure the best possible communication between the ENERCON SCADA Server and the outside world via OPC XML DA, setting up a permanent online connection between the server and the client is recommended. The specific implementation is at the customer's discretion.

Since the refresh rate for online values on the OPC server depends on the speed of the internal data bus, the fastest possible connection between the individual wind energy converters, the ENERCON SCADA Server and other, optional components should be used inside the wind farm.

For more detailed information concerning the requirements for the data connection of the wind farm, refer to the "ENERCON SCADA Datenanbindung des Windparks" (ENERCON SCADA data connection of wind farms) (D0140738) document.

The customer must grant ENERCON Service access through the following port:

Tab. 8: Port to be enabled

Port	Direction	Purpose
6010	Incoming	ENERCON SCADA PDI-OPC



For security reasons, only private IP addresses that have been configured for OPC access in the SCADA system will be unblocked in the internal SCADA firewall. Only the static IP addresses configured in this manner can access the OPC server. Public IP addresses cannot access or be enabled in the internal SCADA firewall.

Software compatibility

ENERCON SCADA version 2.0 or greater (Linux-based operating system)

The ENERCON SCADA Server must support the Linux operating system and must have the performance features required by the ENERCON SCADA software. Especially with existing wind farms, the installed ENERCON SCADA Server may need to be replaced. Any existing wind farm data can be copied to the new server.

Installation location

Preferably, the ENERCON SCADA Server should be located in the wind farm's transmission substation. ENERCON Service IT must have unrestricted access. This will have to be considered particularly during the planning stage.

Safety

ENERCON SCADA PDI-OPC may only be operated if all communication links are secured by a VPN tunnel.

5 Required maintenance

As long as the ENERCON SCADA PDI-OPC is working fault-free, no maintenance is required. The hardware of the ENERCON SCADA Server is included in the visual inspection that is part of the maintenance on the ENERCON SCADA system, provided a maintenance contract or ENERCON PartnerKonzept contract with ENERCON is in place. The ENERCON SCADA PDI-OPC is not part of the ENERCON PartnerKonzept.

Devices installed between the ENERCON SCADA PDI-OPC interface of the ENERCON SCADA Server and the customer's client device are not included in the maintenance provided by ENERCON Service.

6 Scope of supply and services

ENERCON SCADA PDI-OPC is an optional component and is not included in the regular scope of supply of an ENERCON wind farm.

If agreed, the relevant hardware and software is supplied in the form of the ENERCON SCADA Server. Physically, the presence of an ENERCON SCADA PDI-OPC is indicated only by the RJ 45 network connector.

For a detailed description of nodes and items, please refer to the “Datenblatt ENERCON SCADA PDI-OPC” (Data Sheet ENERCON SCADA PDI-OPC) (D0138302).

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Glossary of terms

Array	In computer science, an array is a data structure. Using a field, data of a typically standardised type can be stored in computer memory in an arranged manner. Data access is made possible by using an index.
Client	A client is a computer program that connects to a server in order to make use of services running on the server. Various services can be accessed, depending on the type of client.
ENERCON SCADA METEO	The ENERCON SCADA METEO system serves to acquire and analyse meteorological data in combination with ENERCON SCADA.
Flag	In computer science, a flag is a binary variable or a status indicator. A flag can be used to identify specific states.
Item	Within the context of OPC, the combination of a (physical) value and its properties is called an Item.
Namespace	A namespace is an abstract container providing context for the items (names, or technical terms, or words) it holds. The use of namespaces allows disambiguation of homonym items having the same name (for instance, identical telephone numbers with different area codes).
Node	Within the context of OPC, a folder/directory for structuring the OPC namespace file system. A node can contain both items and additional nodes.
OPC XML-DA	Open Platform Communications, see www.opcfoundation.org . Extensible Markup Language (XML) is used, among other things, for the exchange of data between computer systems, in particular over the Internet. DA means "Data Access".
Protocol	Definition of standards and conventions for trouble-free data transmission between participants in communication.
RTU-C	ENERCON SCADA Remote Terminal Unit Control (RTU-C) is a superordinate closed-loop wind farm control system. RTU-C uses the output values registered at the network connection point to achieve the specified setpoints.
Server	A server is a program that communicates with other programs and can provide special services and data to the latter.
Thread	In computer science, a thread of execution or a processing sequence during execution of a program. Threads enable parallel processing.